# Juveiria Khan, MASc

+1 647 228 6355 | khan.juvi786@gmail.com | linkedin.com/in/juveiria-khan

## Profile Summary

Mechanical Engineering Graduate with hands-on experience in materials engineering, experimental research, and laboratory testing. Skilled in data analysis, process improvement, and developing practical engineering solutions across research and manufacturing environments. Adaptable and detail-oriented professional with strong collaboration skills and the ability to contribute to multidisciplinary teams.

## Education

**York University** — **Sep 2023 – Sep 2025**
*Master of Applied Science in Mechanical Engineering* — *North York, Canada*

- Thesis: Laser-Carbonized Graphene And Lignin Films For Sustainable Transient RFID Electronics
- Oral Presentation: Biodegradable and flexible RFID electronics: Carbon-based eco-friendly materials for a circular economy, *CSME-CFDSC-CSR International Congress, 2025*

**Aligarh Muslim University** — **Aug 2019 – May 2023**
*Bachelor of Technology in Mechanical Engineering* — *Aligarh, India*

- Project: Assesment of Mechanical Properties of Natural And Hybrid Composite
- Oral Presentation: Tensile and Flexural Properties of Woven Bamboo and Jute Reinforced Epoxy Based Hybrid Composites, *CSME/CFD, 2024*

## Skills

**Tools & Equipment:** Laser Processing, Thermal & Mechanical Testing, Polymer Nanocomposites, Polymer Processing (Hot Press, Melt Mixing), Material Characterization, Spectral Analysis, Quantitative Analysis, MATLAB, Microsoft Excel (Advanced), OriginLab

**Languages:** Python, C and C++ (Intermediate Level)

**Soft Skills:** Communication, Team-work, Leadership, Attention to Detail, Problem Solving, Time Management

## Experience

**Graduate Research Assistant** — **Sep 2023 – Sep 2025**
*Polymer and Inorganic Composites, Structures and Surfaces Lab, York University* — *North York, Canada*

- Fabricated thin functional films on polymer substrates using doctor blade coating and operated laser processing systems to develop conductive patterns and optimize processing parameters.
- Worked with a compressible flow exfoliation (CFE) system to synthesize graphene and integrate it into coating formulations for material development.
- Obtained low sheet resistance ($\sim 16\,\Omega/\text{sq}$) on coated samples and evaluated material properties via mechanical & electrical testings.
- Conducted experimental testing and data analysis utilizing scanning electron microscopy, thermogravimetric analysis, Raman spectroscopy, and Fourier transform infrared spectroscopy to evaluate material performance.

**Graduate Teaching Assistant** — **Sep 2023 – Sep 2025**
*York University* — *North York, Canada*

- Courses: Effective Engineering Communication; Engineering Projects: Management, Economics & Safety; Engineering Design Principles; Chemical Dynamics (Lab); Professional Engineering Practice.
- Graded coursework and delivered tutorials, labs, and one-on-one academic support to strengthen student understanding of engineering fundamentals and technical skills.

**Research Intern** — **May 2024 – Aug 2024**
*École de technologie supérieure - ÉTS* — *Montreal, Quebec*

- Worked on a research study related to sorting and recycling of black plastics containing nanomaterials.
- Fabricated graphene-reinforced HDPE and polypropylene nanocomposites using hot press processing and melt mixing.
- Performed FTIR spectroscopy (ATR mode) for material characterization and analysis of polymer–nanofiller interactions.
- Conducted quantitative data analysis using spectral area ratios and developed calibration curves to estimate graphene concentration, ensuring data accuracy and reliability.

**MITACS Research Intern** — **Jun 2022 – Aug 2022**
*Ontario Tech University* — *Oshawa, Canada*

- Fabricated electrospun nanofibrous scaffolds using polymer solutions (PVA, PVDF) under controlled processing conditions.
- Optimized electrospinning parameters to achieve desired fiber morphology and porosity for biomaterial applications.
- Performed mechanical testing (DMA) to evaluate tensile properties, elongation, and structural performance of materials.
- Conducted microscopy-based analysis to assess fiber structure, thickness, and material behavior.